

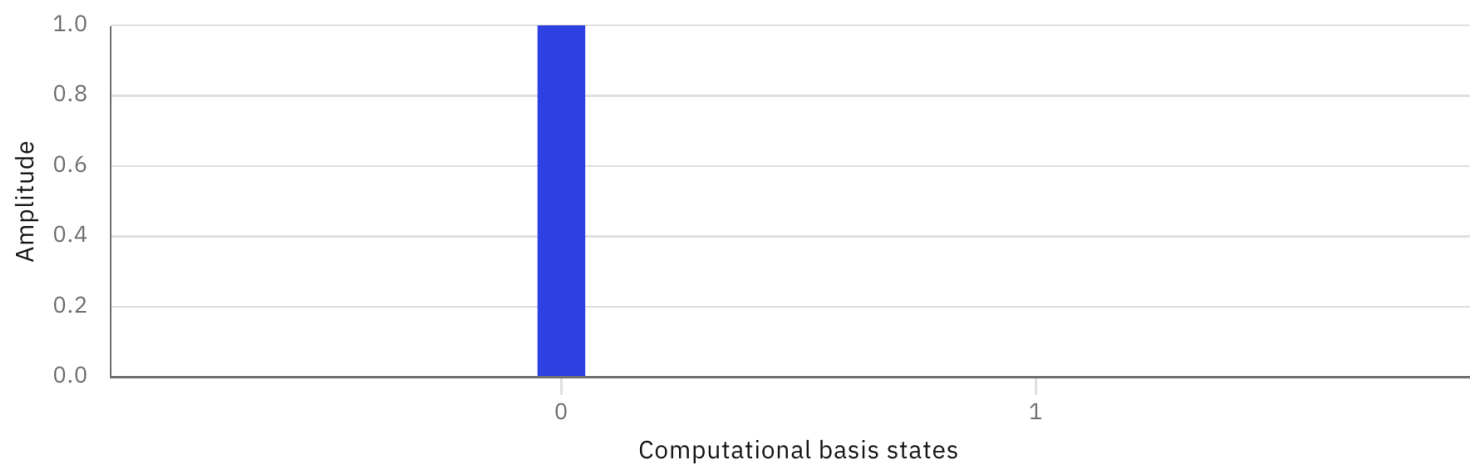
Das NOT-Gate und das Hadamard-Gate

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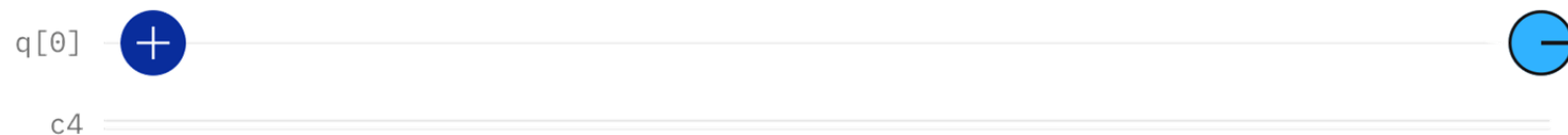
NOT-gate



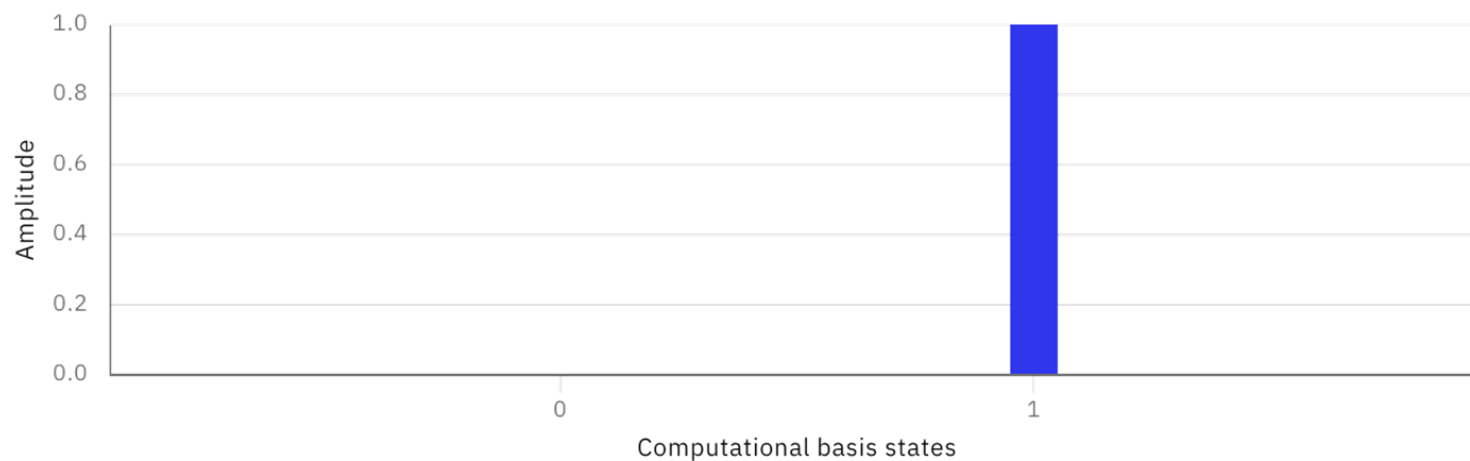
Statevector 



NOT-gate



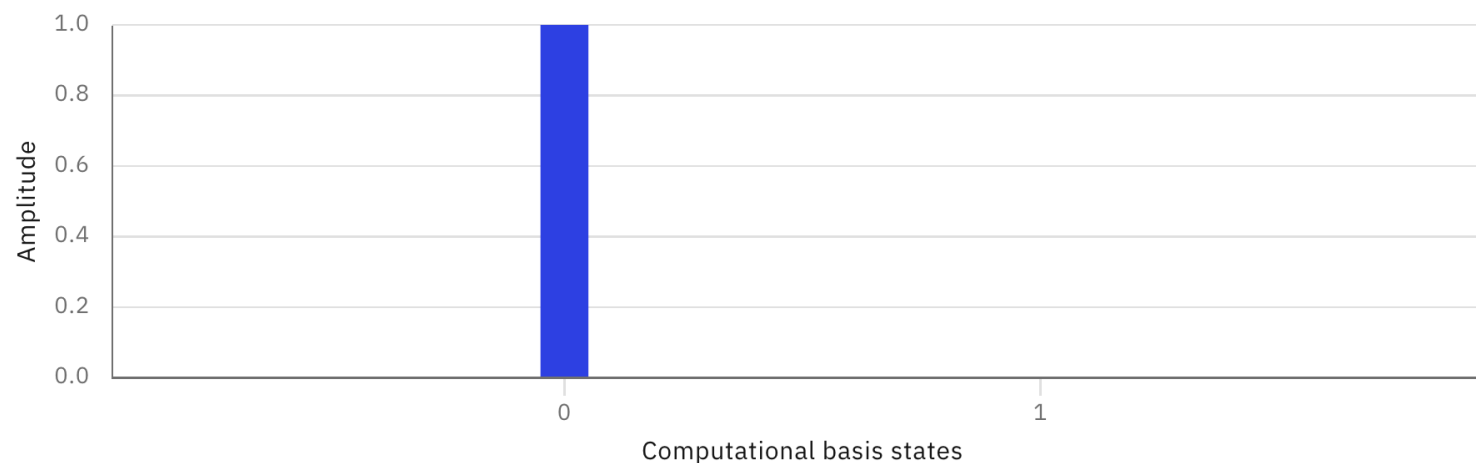
Statevector 



NOT-gate

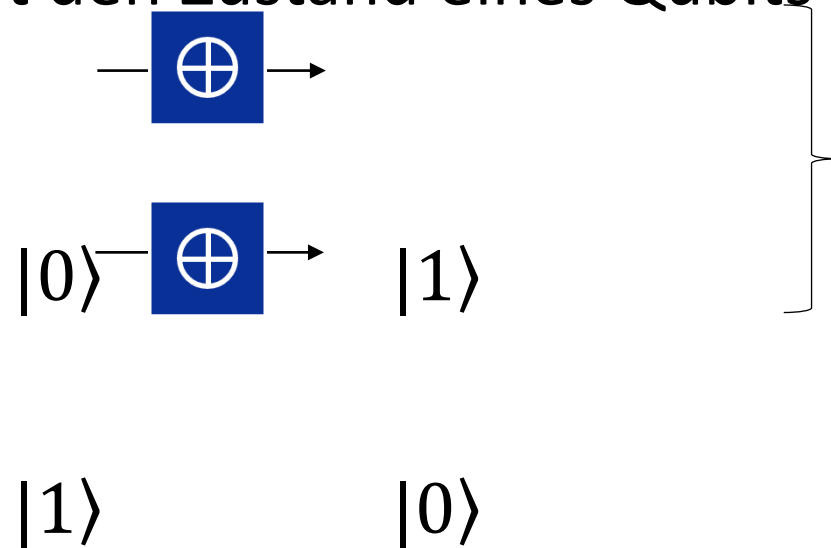


Statevector ▼



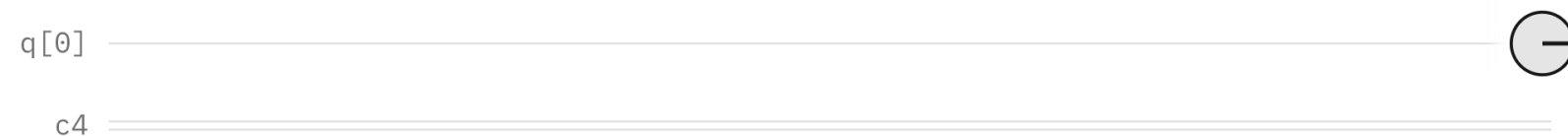
NOT-gate

- Wirkt auf einzelnes Qubit
- Kehrt den Zustand eines Qubits um:

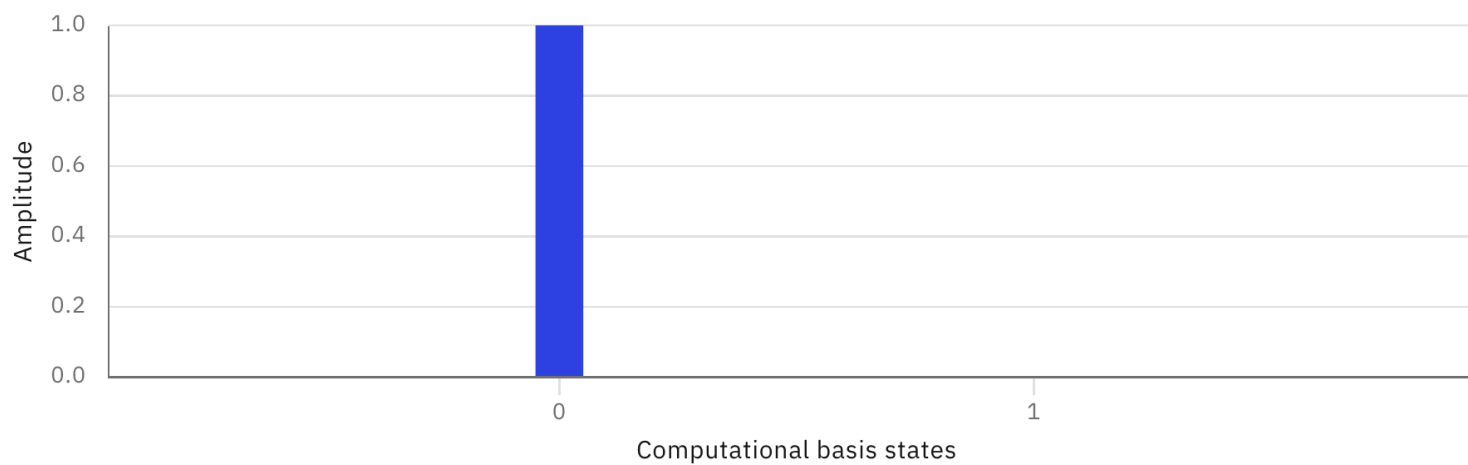


$$\alpha|0\rangle + \beta|1\rangle \text{ --- } \text{NOT gate} \text{ --- } \alpha|1\rangle + \beta|0\rangle$$

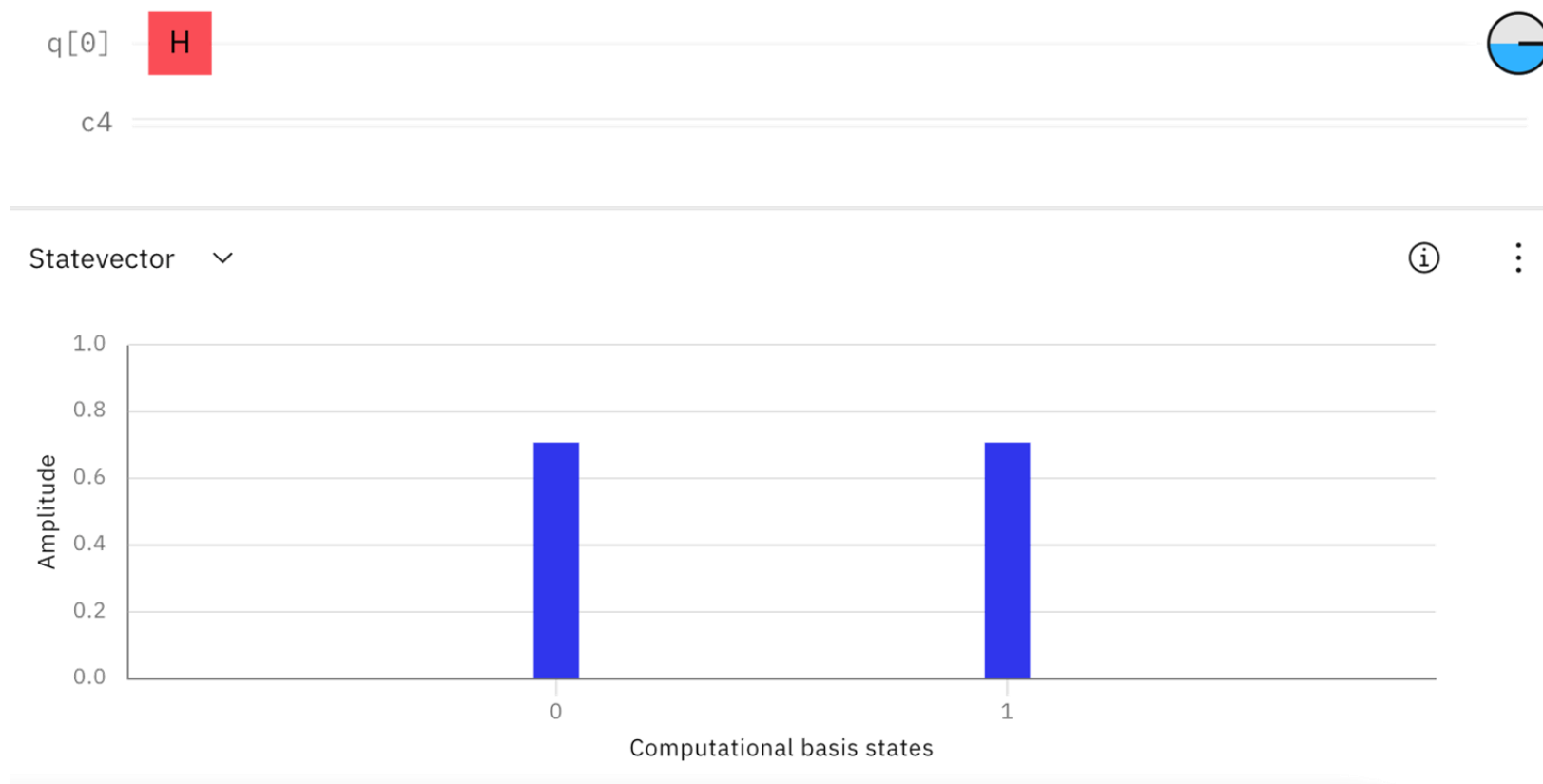
Hadamard-gate



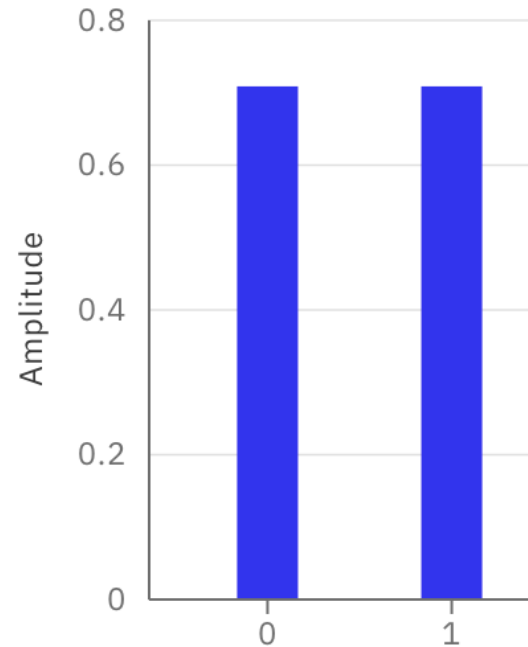
Statevector 



Hadamard-gate



Hadamard-gate



- Zwei Amplituden!
- Qubit ist in Superposition

- Wirkt auf einzelnes Qubit

- Erzeugt **Superposition!**

$$|0\rangle \quad \longrightarrow \quad \boxed{\text{H}} \quad \longrightarrow \quad \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

$$|1\rangle \quad \longrightarrow \quad \boxed{\text{H}} \quad \longrightarrow \quad \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$